AI-assisted loan file preparation

An underwriting copilot, SBA 7(a) first

The tool prepares the file. The underwriter decides.

Team briefing and demo. Runs locally on synthetic data. No real borrower data, no real lender system, nothing deployed.

The problem: preparation crowds out judgment

1. **Read**
   Tax returns, bank statements, financials, ownership, the request

2. **Key and cross-check**
   Do statements add up? Are they recent? Do owners total 100 percent?

3. **Apply the rules**
   SBA SOP 50 10 requirements, one by one

4. **Write the memo**
   Every figure traced back to a document

Hypothesis

If the file arrives prepared, with every figure linked to its source and problems flagged, underwriters spend less time per file and miss fewer problems.

Unproven until measured on real files.

What we are building

Application

Structured intake; tax ID encrypted

Documents

Checked, scanned, facts read with page and quote

Rules

11 SBA SOP 50 10 rules plus consistency checks

Exceptions and memo

Problems flagged; cited memo drafted

Underwriter decides

Only a human can record it

AI extracts and drafts

Calculations and rules are tested code, not model output.

A human decides

The software never approves, declines or ranks a borrower.

Everything is cited and audited

No fact without a page and quote. Append-only, hash-chained log.

Synthetic data only. A human underwriter makes every credit decision.

Who uses it

Processor

Can

Enters the application, uploads documents, builds the file

Cannot

Cannot record a decision

Underwriter

Can

Reviews the file, checks facts against sources, disposes exceptions, decides

Cannot

Cannot administer users

Viewer

Can

Reads files

Cannot

Cannot change anything

Admin

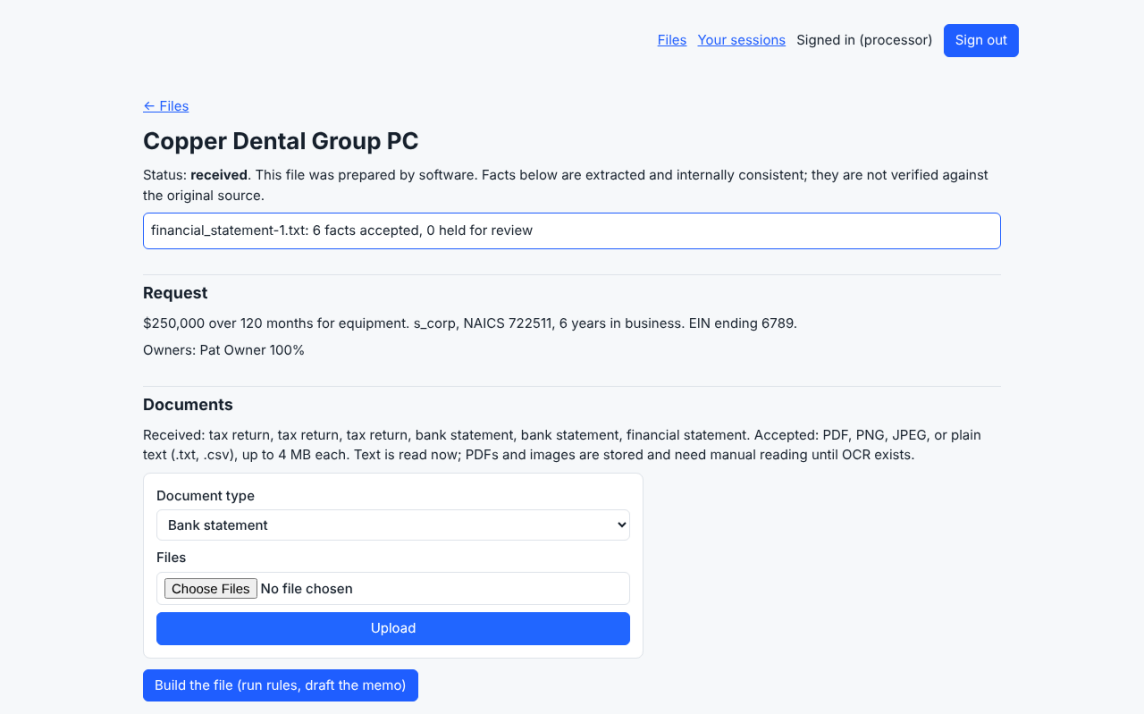
Can

Users, roles, sessions, security events, the lender's own sign-in setup

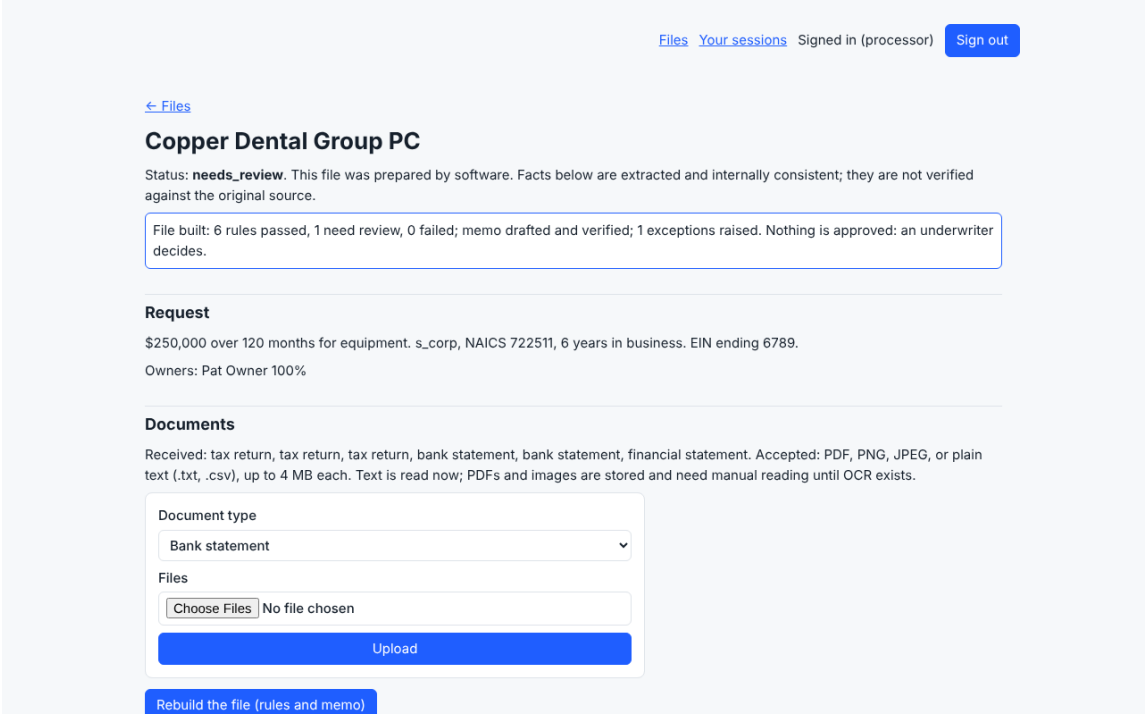
Cannot

Cannot record a decision

Live: documents in, file built

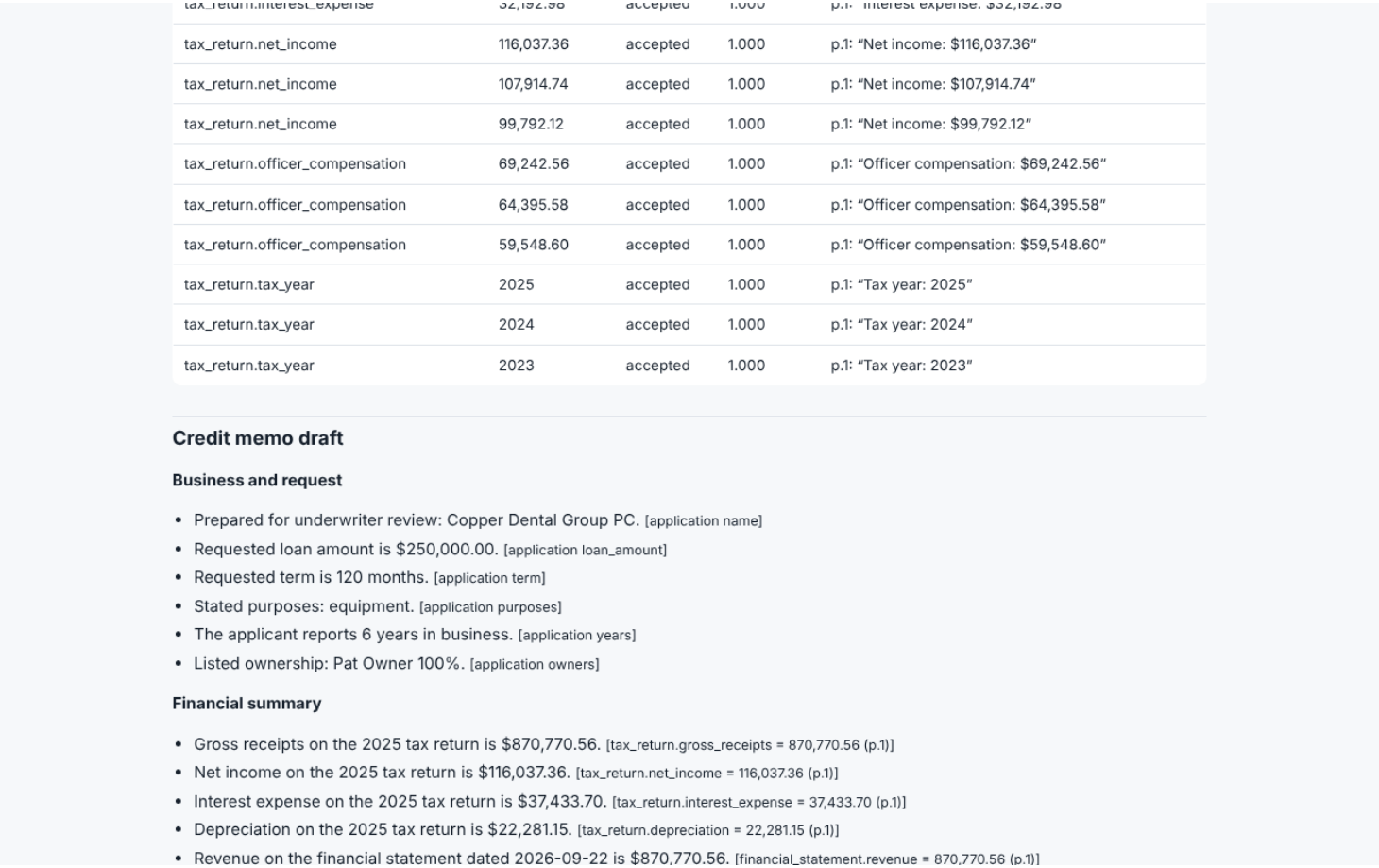


The processor uploads the packet. Each file is checked for type, content and malware before it is kept.



One click runs the rules and drafts the memo. The message states that nothing is approved.

Live: the underwriter's file



tax_return.net_income	116,037.36	accepted	1.000	p.1: “Net income: $116,037.36”
tax_return.net_income	107,914.74	accepted	1.000	p.1: “Net income: $107,914.74”
tax_return.net_income	99,792.12	accepted	1.000	p.1: “Net income: $99,792.12”
tax_return.officer_compensation	69,242.56	accepted	1.000	p.1: “Officer compensation: $69,242.56”
tax_return.officer_compensation	64,395.58	accepted	1.000	p.1: “Officer compensation: $64,395.58”
tax_return.officer_compensation	59,548.60	accepted	1.000	p.1: “Officer compensation: $59,548.60”
tax_return.tax_year	2025	accepted	1.000	p.1: “Tax year: 2025”
tax_return.tax_year	2024	accepted	1.000	p.1: “Tax year: 2024”
tax_return.tax_year	2023	accepted	1.000	p.1: “Tax year: 2023”

Credit memo draft

Business and request

- Prepared for underwriter review: Copper Dental Group PC. [application name]
- Requested loan amount is $250,000.00. [application loan_amount]
- Requested term is 120 months. [application term]
- Stated purposes: equipment. [application purposes]
- The applicant reports 6 years in business. [application years]
- Listed ownership: Pat Owner 100%. [application owners]

Financial summary

- Gross receipts on the 2025 tax return is $870,770.56. [tax_return.gross_receipts = 870,770.56 (p.1)]
- Net income on the 2025 tax return is $116,037.36. [tax_return.net_income = 116,037.36 (p.1)]
- Interest expense on the 2025 tax return is $37,433.70. [tax_return.interest_expense = 37,433.70 (p.1)]
- Depreciation on the 2025 tax return is $22,281.15. [tax_return.depreciation = 22,281.15 (p.1)]
- Revenue on the financial statement dated 2026-09-22 is $870,770.56. [financial_statement.revenue = 870,770.56 (p.1)]

Cited

Every fact carries document, page and the quoted passage.

Checked

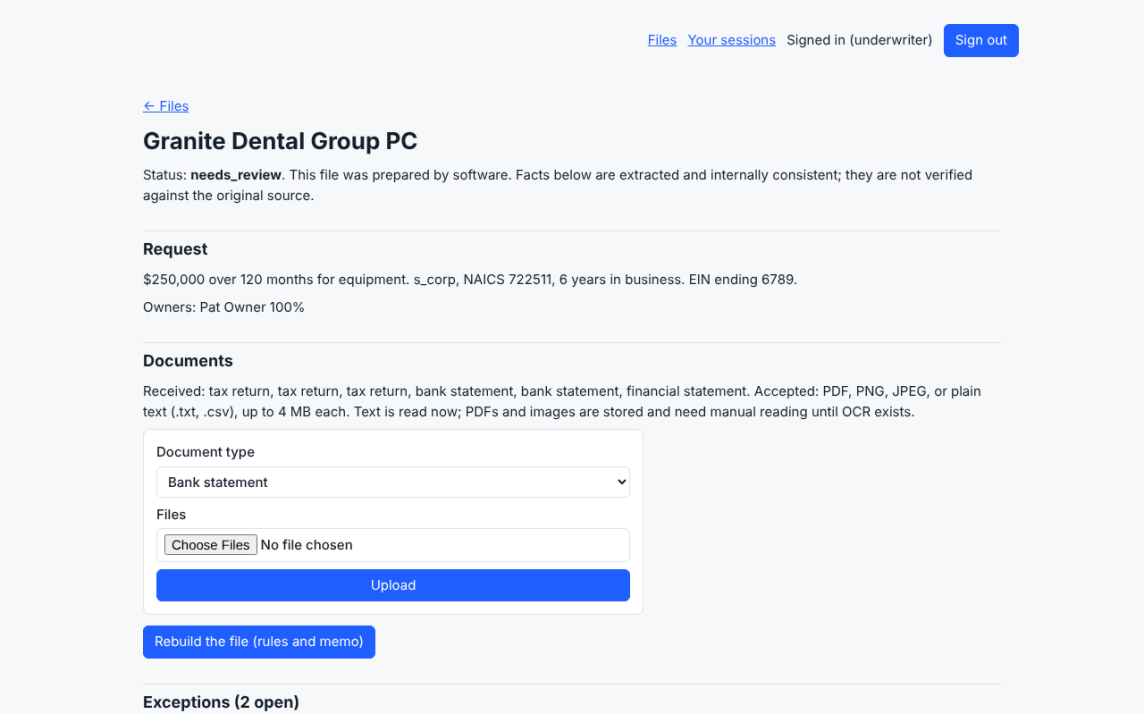
Memo numbers are verified against their cited source.

Honest

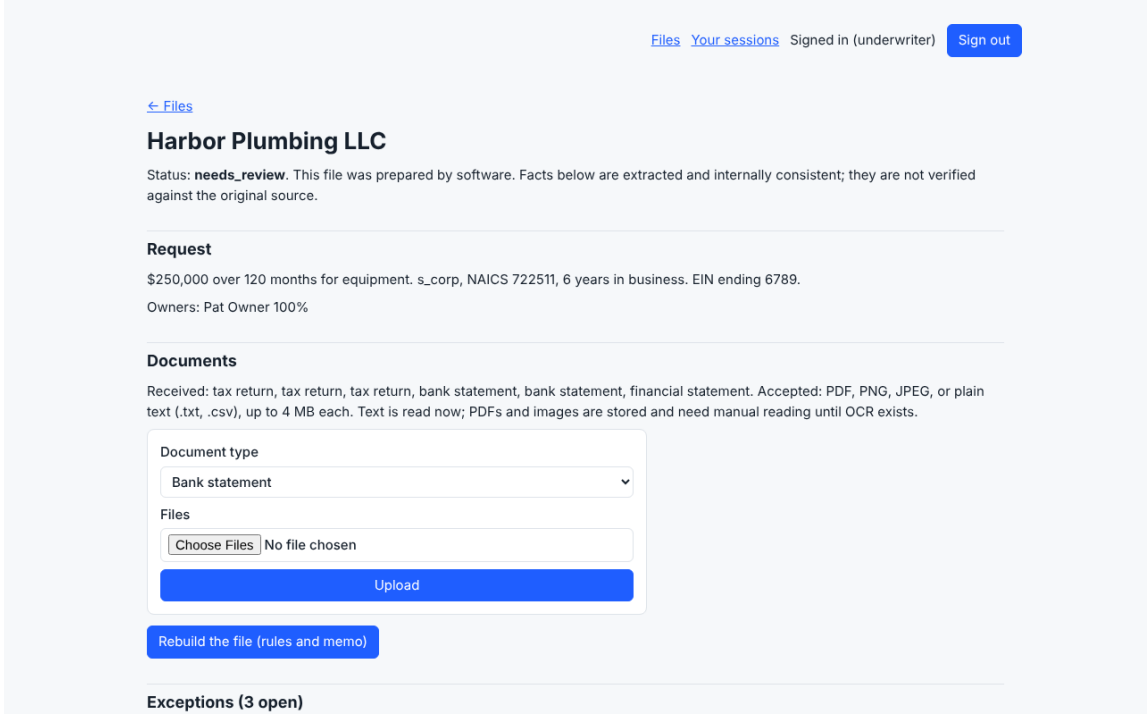
The tool lists what it did not evaluate.

Synthetic data only. A human underwriter makes every credit decision.

Live: problems are caught

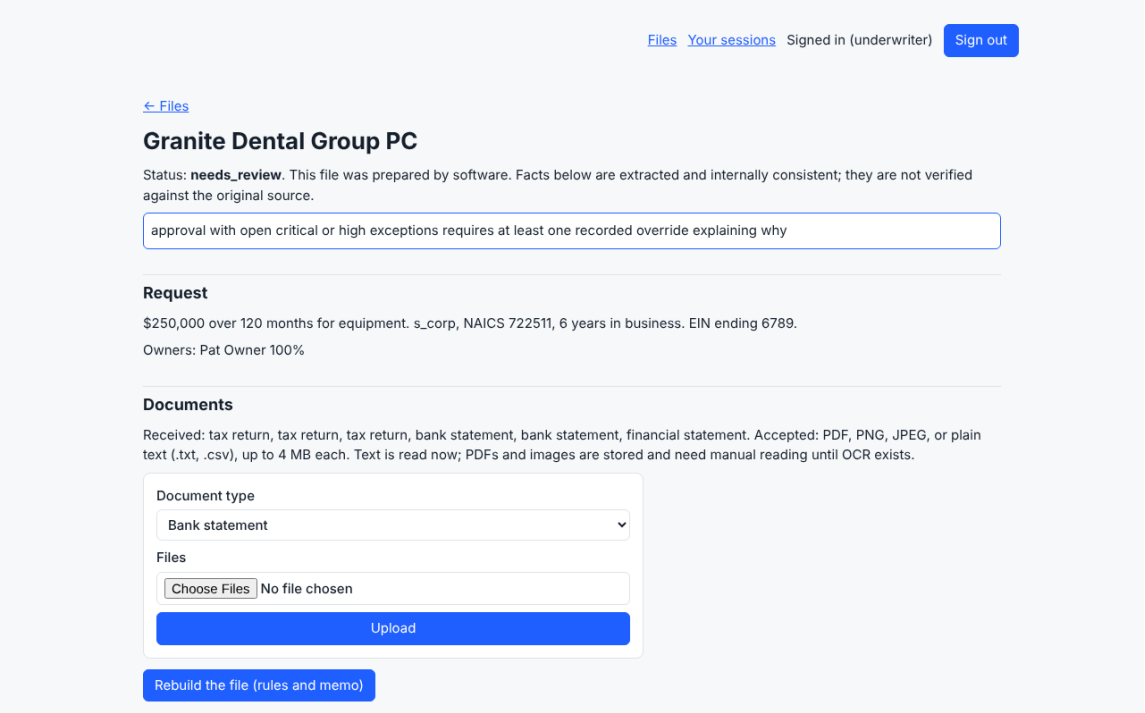


Statements that do not add up raise a high-severity exception.

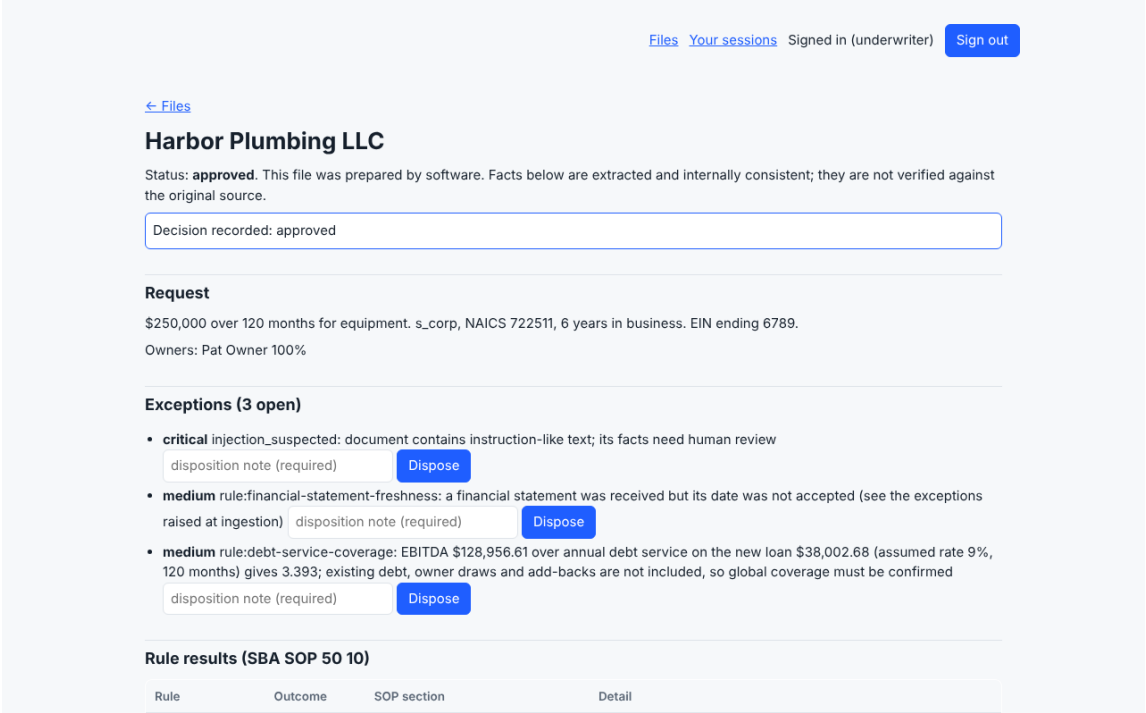


A document tried to instruct the system. It was treated as data, flagged critical, and its facts held for a human.

Live: the human gate



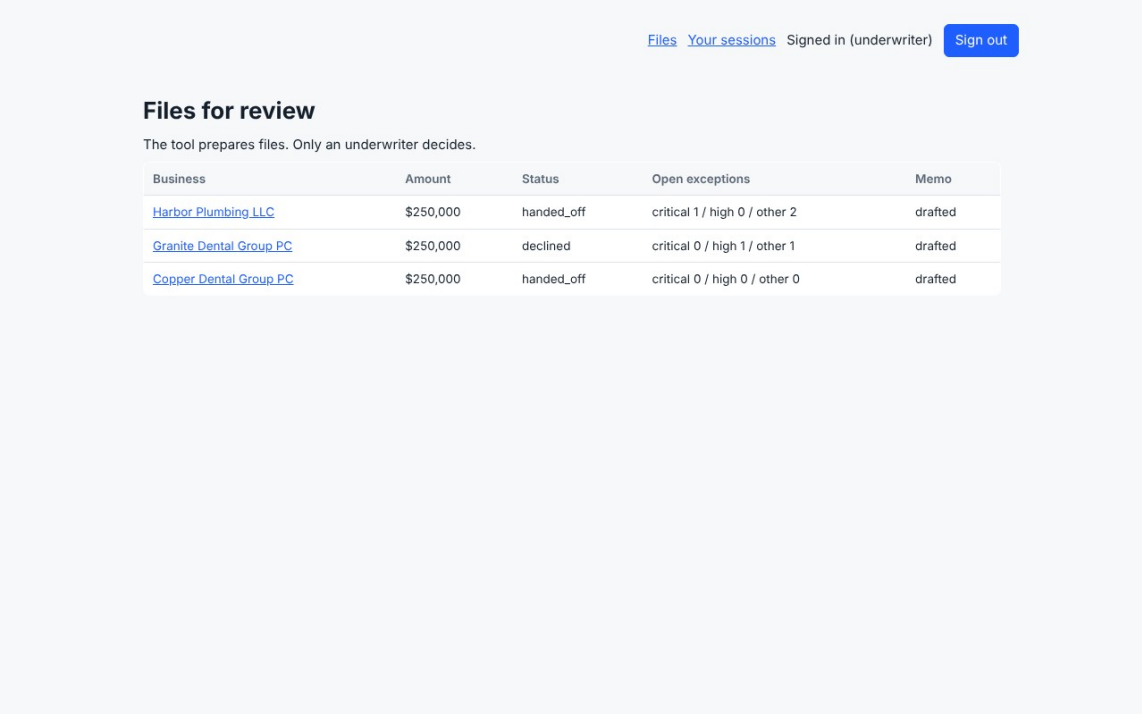
Approve past an open high exception with no override: refused, by the database.



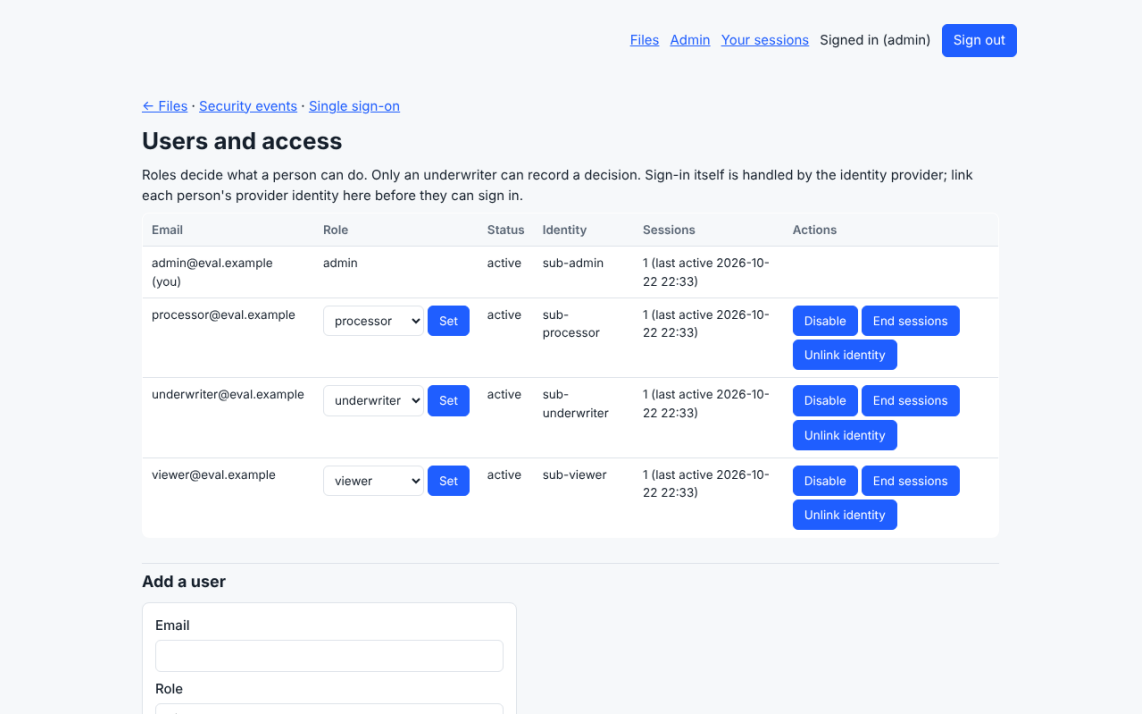
With a written override from an underwriter: recorded with the reason.

Synthetic data only. A human underwriter makes every credit decision.

Live: after the decisions, and admin



Two files handed off to the mock servicing adapter, one declined.



Admin: users, roles and sessions. Every change is audited.

How it is built to be trusted

Lender isolation

Forced row-level security on every table. The app role is not an owner and cannot bypass it.

Human-only decisions

Enforced by database rules, not only screens. Decisions are append-only.

Evidence

No fact without a page and quote. Model confidence never decides acceptance.

Safe documents

Name must match content; markup and active PDFs refused; malware scan fails closed.

Sign-in and access

OpenID Connect, four roles, revocable sessions, every change audited.

Audit

Hash-chained, append-only ledger the database can verify.

What we have measured

454

automated tests pass, including real-database tests

40 of 40

browser walkthrough steps pass: four roles, three files

102

facts in the run, every one with a source passage

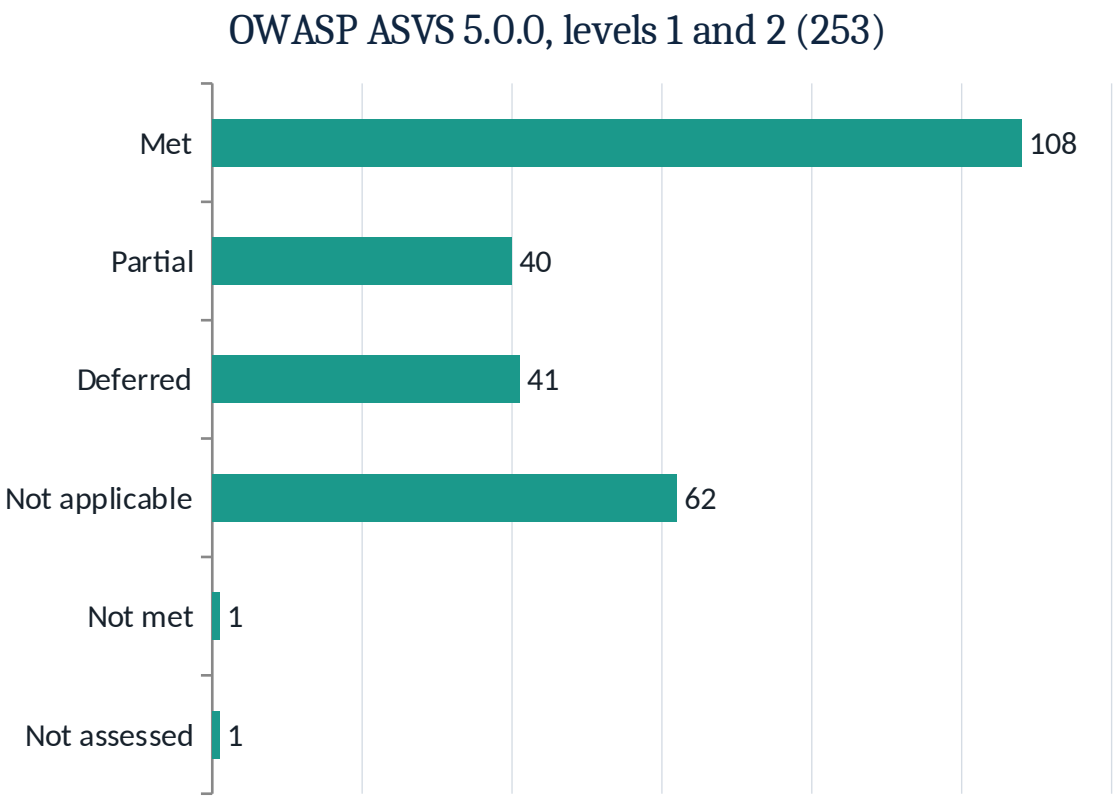
0

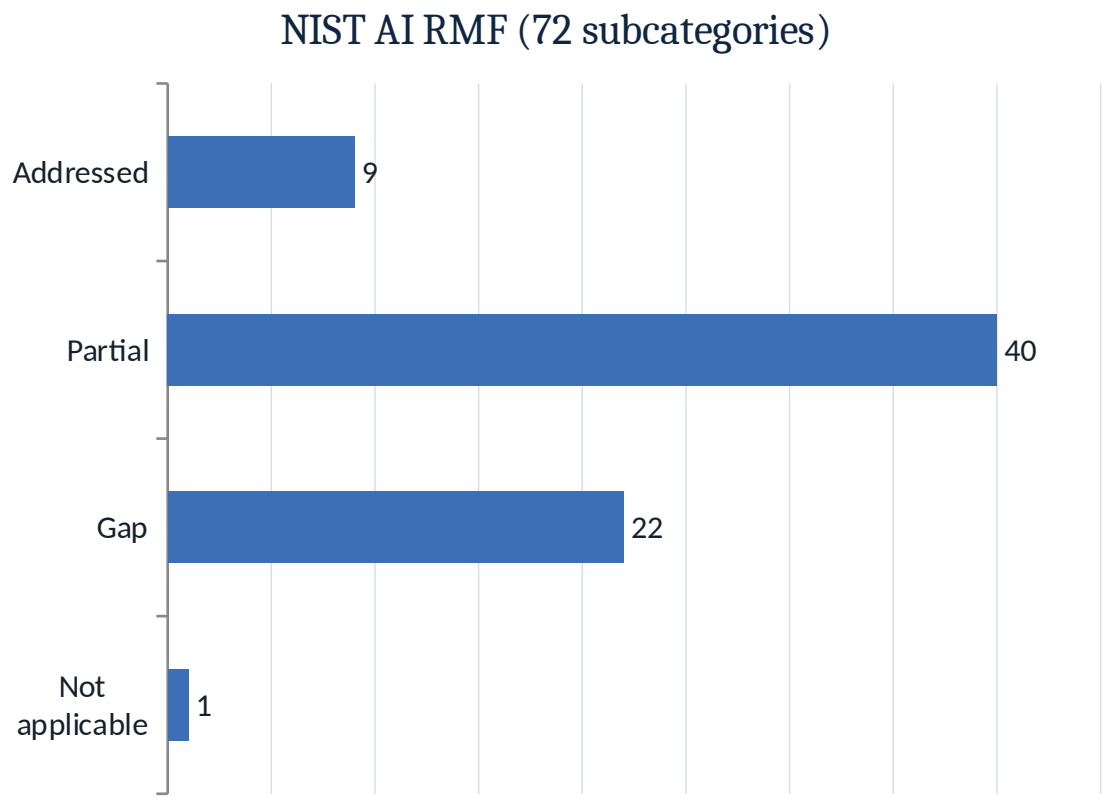
falsely accepted fields in 4,030 (synthetic, stand-in model)

Also measured: 87.8 percent of fields accepted and correct, 8.4 percent not extracted (image-only files, no OCR). Every planted problem class was caught in 100 percent of packets; two classes also over-flagged.

Labels: measured on synthetic data. Not measured: accuracy on real documents, time saved.

Our own reading of the security and AI standards





Self-assessed. 72 percent of the 150 applicable ASVS requirements are met. The gaps are listed in the repository with owners.

What it does not do yet

- No real model provider or OCR; a stand-in model reads printed labels
- No real documents or lenders have been tested
- No identity provider chosen; sign-in tested with a stand-in
- Not deployed; hosting is undecided

- Not evaluated: eligibility, collateral, tax transcripts, personal financial statements, credit reports, global cash flow
- Debt service uses only the new loan and a placeholder 9 percent rate
- 504 and SBA Express are not encoded
- No third-party security test, SOC 2, vendor review or counsel review

How we compare, honestly

Capability	Leading products (public claims)	This tool today
Scale, integrations, SBA forms and E-Tran	Yes (nCino, Abrigo, Baker Hill)	**Behind: none**
SOC 2 Type 2, ISO 27001, penetration test	Listed by several vendors	**Behind: none yet**
Fact-level citations, checked memo numbers	Described as sourced; method not described	**Built and tested**
Human decision enforced by the database	Not described on pages read	**Built and tested**
Accuracy published with its failures	Claims without method	**Published, but synthetic only**

No head-to-head test has been run. We will pre-register targets, test on the same files, and publish the result either way.

The next 90 days

Days 1 to 30

Learn and baseline

- Lender and processor conversations
- Timed manual baseline with two underwriters
- Choose identity provider
- Decide hosting

Gate: requirements and a measured baseline

Days 31 to 60

Make it real

- One real model provider, after reviewing its data terms
- OCR for scans and images
- Rerun the evaluation, publish failures

Gate: go or no-go

Days 61 to 90

Pilot

- Partner's policy inputs and debt schedule
- External penetration test; counsel review
- Redacted, consented files; time both flows

Gate: measured comparison; continue or stop

Decisions we need from the team

1. Which lending lines and volumes first, and what outcome defines success?
2. Who can arrange lender conversations and two underwriters for the timed baseline?
3. Hosting for a pilot (we recommend AWS containers; Vercel for testing only) and who approves vendor risk?
4. Which model provider to evaluate, and approval for external calls with synthetic data first?
5. Which identity provider?
6. Who reviews the rules against SOP 50 10 and the list for counsel?

Demo plan: 15 minutes

Min	Do	Point to make
1	**Sign in as Underwriter**	Roles come from our database; sessions revocable; sign-ins audited
1	**Open the queue**	Every file is needs review; the software never approves
2	**Open a flagged file**	Instruction-like text held for a human, not obeyed
3	**Rules, facts, memo**	11 SBA rules with SOP sections; every fact cited; memo numbers checked
2	**Try Approve, then add an override**	Refused first, recorded with a written reason; enforced by the database
1	**Audit trail and handoff**	Append-only, hash-chained; only the approved file reached the mock adapter
3	**Processor: apply, upload, build**	Intake to prepared file in a few clicks; tax ID shows last four only
1	**Admin screens**	Users, roles, sessions, security events

The tool prepares the file.

The underwriter decides.

Reference: workspace/docs/TEAM_GUIDE.md. Demo: demo/walkthrough/WALKTHROUGH.html. Nothing here is legal advice.